SAR-RAG — Independent Verification Report

Representation-State Adaptive RAG | commit 5cb4cd7dc85f9c4bd87f22cfa840494d98b6a9d5 | **Zero paid Groq calls were made during this audit**

How to read this report

Label	Meaning
REPRODUCED FACT	Re-derived from source code or recorded artifacts during this audit
CALIBRATION DIAGNOSTIC	Training-only calibration output; not an evaluation result
ENGINEERING PILOT	From the 2-question QASPER pilot; not inferential
CONFIRMATORY RESULT	From a completed held-out run — none exist in this repository
PENDING	Not executed
REVIEWER INFERENCE	Auditor judgement, flagged as such
LIMITATION	A constraint on what can be concluded

1. Headline findings

#	Finding	Label
1	**No result artifact in this repository was produced by the current code.** No run_fingerprint.json and no checkpoints/ directory exist anywhere under output/. The pilot lacks complete_evidence_recovery, gold_evidence_spans, candidates_examined, retrieval_rounds, stop_reason, trace_summary, provisional_evidence and api_attempts. Corrections #12, #13, #15 and #16 were in source but never executed.	REPRODUCED FACT
2	**The frozen calibration reproduces bit-for-bit.** All 32 proposed trials, 6 baseline trials, sample IDs, partitions, strata and source hashes are identical, before and after defect remediation. No threshold changed.	REPRODUCED FACT
3	**Three implementation defects found and fixed**, one falsifying a claimed correction: peak working set was always null on 64-bit Windows (truncated HANDLE); retrieval recall still returned 1.0 with no gold evidence in a second live path; the baseline reported final_k as candidates examined.	REPRODUCED FACT
4	**A single-view baseline currently beats the full routed system.** On retrieval-only screening over fresh training examples, paragraph_only beats the full four-view routed system on recall, complete-evidence, precision, context tokens AND candidates examined simultaneously. This is the most serious threat to the paper's contribution (see 6b).	REPRODUCED FACT
5	**Two named components are inert; the second view is near-inert.** no_novelty and no_similarity_gap reproduce the full system to four decimal places. Two views are selected for 16.7% of queries and add unique gold evidence in only 3.3%.	REPRODUCED FACT
6	**But at a matched context budget, SAR-RAG does win.** At ~1,300 tokens it beats Traditional k=10 by +0.056 recall and +0.116 complete-evidence. The efficiency claim must be framed budget-matched rather than k-matched.	REPRODUCED FACT
7	**Leakage is smaller than the shipped audit implies.** Only the 60 calibration examples were ever fitted; that check did not exist. They share 0 example IDs, 0 source IDs and 0 document IDs with either evaluation split.	REPRODUCED FACT

2. Confirmatory results

[PENDING] **No confirmatory experiment has been run.** Every cell below is pending and must not be filled from pilot or calibration data.

Table	QASPER test (n=1,451)	MuSiQue dev (n=4,834)	Status
Main per-dataset results	PENDING	PENDING	requires paid run
Answerability and abstention	PENDING	PENDING	requires paid run
Statistical inference	PENDING	PENDING	requires paid run
Efficiency and cost	PENDING	PENDING	rates unverified; cost undefined
Representation-state analysis	PENDING	PENDING	retrieval-only screening available
Adaptive-k analysis	PENDING	PENDING	retrieval-only screening available
Ablation results (end-to-end)	PENDING	PENDING	15/20 screened offline
Robustness / subgroup	PENDING	PENDING	requires paid run
Failure-category counts	PENDING	PENDING	requires paid run

3. Calibration reproduction

Comparison	Differing fields	Result
v0 original frozen vs v1 reproduction (unmodified code)	0	IDENTICAL
v1 reproduction vs v2 (after defect fixes)	0	IDENTICAL
Thresholds changed by remediation	0	none

[CALIBRATION DIAGNOSTIC] Held-back training-selection partition (24 examples). These are **not** test results and establish no superiority.

Metric	SAR-RAG	Traditional	Difference
Evidence recall	0.7917	0.8958	-0.1042
Complete-evidence recovery	0.6875	0.8750	-0.1875
Evidence precision	0.1719	0.1127	+0.0592
Mean context tokens	1,265	1,833	-31.0%
Mean final k	14.67	17.33	-2.66
Mean selected representations	1.125	1.000	+0.125

4. Leakage audit

[REPRODUCED FACT] 72 distinct QASPER question strings overlap train and test, touching 163 test examples (11.23%). All are generic NLP-paper templates and **0 share a paper** with their training counterpart. This is template repetition, not answer or source leakage — but it must never be described as zero leakage.

Calibration sample (60 fitted examples) vs	QASPER test	MuSiQue dev
Example-ID overlap	0	0
Source / paper-ID overlap	0	0
Document-ID overlap	0	0
Distinct question templates shared	1	0
Evaluation examples touched	2 (0.138%)	0 (0.0%)

[LIMITATION] Exact matching cannot rule out semantic paraphrase, prior benchmark contamination, or model pretraining exposure. The last applies equally to both arms, so it does not bias the paired comparison — only absolute scores.

5. Defect remediation

ID	Class	Sev	Title	Status
D01	a	high	Retrieval recall returned 1.0 when the benchmark declares no gold evidence	FIXED
D02	a	high	Process peak working set silently recorded as null on 64-bit Windows	FIXED
D03	a	medium	Traditional baseline reported final_k as candidates examined	FIXED
D04	a	medium	Calibration retained every sampled corpus for the whole search	FIXED
D05	a	medium	Empty analysis tables were silently not written	FIXED
D06	a	medium	Figure writing selected an interactive matplotlib backend	FIXED
D07	a	low	Research tests could not run in the project virtual environment	FIXED
D08	b	high	RRF threshold is described as a normalized-RRF quantity but is a fraction of total probe mass	DOCUMENTED NOT CHANGED
D09	b	medium	Calibration selection rule applies an undocumented second recall filter	DOCUMENTED NOT CHANGED
D10	b	medium	Partition described as 70/30 is 60/40	DOCUMENTED NOT CHANGED
D11	b	low	Representation tie-break uses probe words, not probe tokens	DOCUMENTED NOT CHANGED
D12	b	low	Pause patience documented as two batches, frozen at three	DOCUMENTED NOT CHANGED
D13	c	medium	Representation granularity partially confounds the context-token claim (severity REVISED DOWN after measurement)	CONTROL ADDED AND MEASURED
D14	c	high	Complete-evidence recovery is scored on questions declared unanswerable	OPEN REQUIRES DECISION
D15	c	medium	Primary outcomes have very different valid paired n	OPEN DISCLOSE
D16	c	medium	Second representation is selected in a small minority of queries	DIAGNOSTIC ADDED
D17	c	medium	Chunk-level rank metrics are not comparable across arms	OPEN DISCLOSE
D18	c	low	One invalid claim downgrades an entire answer to INSUFFICIENT	OPEN DISCLOSE
D19	a	low	Confirmatory figure captioned as a pilot	OPEN COSMETIC
D20	c	low	Embedding revision pin is inert on the local snapshot path	OPEN DISCLOSE
D21	c	high	A single-view baseline outperforms the full routed system on retrieval-only screening	OPEN CRITICAL
D22	c	high	The novelty and similarity-gap components have no measurable effect	OPEN CRITICAL
D23	c	medium	The RRF constant is nearly irrelevant	OPEN DISCLOSE
D24	c	medium	The second representation contributes almost nothing	OPEN CRITICAL
D25	c	low	Probe depth 3 is not the best probe depth on training data	OPEN DISCLOSE
D26	c	medium	Thirty percent of queries terminate at the final-chunk safety cap	OPEN DISCLOSE

Class a = direct implementation bug, b = documentation/formula mismatch, c = methodological decision requiring a new experimental version.

6. Test suite

	Pre-audit	Post-audit
Tests	23	73
Interpreter	Anaconda (different environment)	.venv (the research environment)
Primary pipeline under frozen controls	not covered	covered
Real MiniLM integration	not covered	covered
Harness fingerprint / checkpoint / resume	not covered	covered
Systems measurement fields	not covered	covered

*The pre-audit suite asserted the two headline contracts (probe depth is not final k, context budget never exceeded) against the **legacy** multi-call system, and its only primary-pipeline test disabled every adaptive gate.*

6b. Retrieval-only ablation screening

[REPRODUCED FACT] Zero Groq calls. Fresh deterministic stratified sample of 60 official TRAINING examples (seed 20260804, distinct from the calibration seed so screening does not reuse threshold-selection examples). 43 of 60 carry gold evidence and are scored. **[LIMITATION]** Retrieval-only: no answer quality, latency or cost, and no statistical inference. This is screening evidence for deciding what deserves paid evaluation, not a result.

Condition	Recall	Compl.ev	Prec.	Ctx tok	final k	Cand.	2-view%
full	0.7054	0.5814	0.1595	1312.7	15.15	33.4	16.7
sentence_window_only	0.6996	0.5349	0.2043	1321.3	15.77	22.6	0.0
paragraph_only	0.8004	0.6512	0.1751	1281.2	15.27	22.9	0.0
semantic_only	0.7171	0.5581	0.1492	1358.7	16.43	24.2	0.0
hierarchical_only	0.3876	0.2326	0.0722	1408.9	14.18	23.8	0.0
all_representations_no_rrf	0.5950	0.3953	0.1375	1056.2	11.00	38.6	0.0
rrf_probe_only_no_adaptive	0.4671	0.3256	0.3566	303.5	3.33	12.0	16.7
no_novelty	0.7054	0.5814	0.1595	1312.7	15.15	33.4	16.7
no_similarity_gap	0.7054	0.5814	0.1595	1312.7	15.15	33.4	16.7
no_post_retrieval_deduplication	0.7054	0.5814	0.1602	1378.5	16.10	33.4	16.7
select_one_representation	0.6938	0.5814	0.1598	1329.0	15.40	32.3	0.0
probe_depth_5	0.7519	0.6279	0.1577	1330.7	14.80	39.0	10.0
rrf_k0_1	0.7054	0.5814	0.1608	1303.4	15.15	33.4	16.7
rrf_k0_10	0.7171	0.5814	0.1610	1309.6	15.12	33.3	18.3
rrf_k0_60	0.7171	0.5814	0.1610	1309.6	15.12	33.3	18.3

Traditional baseline	Words	Recall	Compl.ev	Prec.	Ctx tok	final k
sliding_window_k3	98.4	0.4535	0.2791	0.3256	469.2	3.00
sliding_window_k5	98.4	0.5504	0.3721	0.2419	760.9	5.00
sliding_window_k10	98.4	0.6492	0.4651	0.1628	1293.0	9.67
sliding_window_k20	98.4	0.7733	0.7209	0.1209	1896.4	15.78
sliding_window_matched_k3	67.8	0.4593	0.3256	0.3876	289.3	3.00
sliding_window_matched_k5	67.8	0.4981	0.3488	0.2884	483.3	5.00
sliding_window_matched_k10	67.8	0.6240	0.4651	0.2023	953.7	10.00
sliding_window_matched_k20	67.8	0.7674	0.6744	0.1392	1794.8	19.13

Five findings that change the assessment

#	Finding
F1	**A single-view baseline beats the full routed system.** paragraph_only beats full on every retrieval outcome at once: recall +0.095, complete-evidence +0.070, precision +0.016, fewer context tokens, 31% fewer candidates examined. This is the most serious threat to the paper's contribution.
F2	**Two named components are inert.** no_novelty and no_similarity_gap reproduce full to four decimal places on every metric. At the frozen thresholds neither rule ever binds.
F3	**At a matched context budget, SAR-RAG does win.** At ~1,300 tokens: recall 0.7054 vs 0.6492 (+0.056) and complete-evidence 0.5814 vs 0.4651 (+0.116) against Traditional k=10. The efficiency claim must be framed budget-matched, not k-matched.
F4	**Auditor self-correction.** The granularity confound was initially rated HIGH from config parameters (2.7x). Measured realized ratio is 1.73x on QASPER and 1.44x on MuSiQue, because sliding windows stop at section boundaries. Severity revised to MEDIUM. It does not explain away F3.
F5	**The second representation contributes almost nothing.** Two views on 16.7% of queries (QASPER 30%, MuSiQue 3.3%); adds unique gold evidence in 20% of those, i.e. 3.3% of all queries. select_one_representation loses only 0.0116 recall and zero complete-evidence.

Components that ARE supported: adaptive retrieval (+0.238 recall over probe-only), RRF selection over no selection (+0.110), post-retrieval deduplication (removes ~1 chunk and 66 tokens with zero recall loss). **Also:** 36.7% of queries stop on a safety cap; the RRF constant is nearly irrelevant across {1,3,10,60}; probe depth 5 outperforms the frozen depth 3 and must NOT be silently adopted.

7. Review decision

Question	Answer
Implementation matches intended pipeline?	Substantially yes; one formula misstatement (D08) plus documented deviations
Exactly one primary LLM call?	Yes — verified structurally and end to end
Final k genuinely adaptive?	Yes in mechanism; scale behaviour PENDING
Selection based on RRF complementarity?	Yes in mechanism; published formula is wrong; second view rare
Deduplication after retrieval?	Yes — accepted IDs equal provisional IDs
Traditional a fair baseline?	Fair on the answer side; efficiency confounded by granularity
Calibration isolated from evaluation?	Yes
Evidence of leakage?	No source/document/example leakage; template repetition disclosed
Metrics correctly calculated?	Yes — 22/22 against hand computation
Tests paired and corrected?	Yes — paired bootstrap, randomization, exact McNemar, Holm
Improves answer quality?	PENDING — unknown, never demonstrated
Improves evidence recall?	No on current evidence (-0.104 on calibration)
Improves complete-evidence recovery?	No on current evidence (-0.187 on calibration)
Improves precision / context efficiency?	Yes — at a matched ~1,300-token budget: +0.056 recall, +0.116 complete-evidence
Meaningful Pareto improvement?	Vs fixed-k Traditional at matched budget: plausibly. Vs a paragraph_only single view: no
Which components are supported?	Supported: adaptive retrieval, RRF over no-selection, dedup. Not supported: novelty, similarity gap, second view, RRF constant
Ready for Q2 submission?	**Not yet** — no confirmatory result exists

8. Claims that must be weakened or corrected

#	Required correction
1	Rewrite the RRF formula. The selector applies a **second** normalization over total aligned probe mass, so the frozen 0.2437 is a share of that mass, not a raw NRRF value.
2	Never use 'training-free' unqualified. The repository is already correct; keep it.
3	Never describe normalized RRF as a calibrated probability.
4	Do not call QASPER leakage zero. State: 72 repeated generic templates, 163 test examples, 0 shared papers, 0.14% calibration-level channel.
5	Report the efficiency comparison budget-matched, not k-matched, and disclose the measured 1.44-1.73x granularity difference.
6	Do not present 'one or two representations' as a general mechanism. Two views on 16.7% of queries, unique gold added in 3.3%, and essentially QASPER-only.
7	**Do not claim novelty or the similarity-gap rule as contributing components.** Both are provably inert at the frozen thresholds.
8	**Do not claim the four-view design is necessary.** paragraph_only currently dominates the full system; hierarchical_only is by far the weakest view.
9	Do not describe final k as unconstrained while 36.7% of queries stop on a safety cap.
10	Do not adopt probe depth 5 from screening; it needs a fresh training-only calibration.
11	Do not report a single n = 6,285 beside all four primary outcomes. Valid paired n is 3,789 / 5,645 / 5,645 / 6,285.
12	Keep chunk-level P@k, MRR and nDCG strictly secondary; they are granularity-sensitive.
13	Do not present the three-calls-per-question study cost as the method's cost.
14	Do not label pilot figures as final results.

9. Current defensible conclusion

SAR-RAG is a model-fine-tuning-free, training-calibrated adaptive retrieval method. Training calibration indicates higher evidence precision and reduced context use, but lower evidence recall and complete-evidence recovery than a static-k Traditional RAG baseline. Its overall quality-efficiency advantage remains unproven pending complete held-out evaluation, registered ablations, systems measurements and manual failure analysis.

[REVIEWER INFERENCE] **Three auditor additions**, all from retrieval-only screening on training data:

#	Addition
1	The efficiency comparison must be **budget-matched, not k-matched**. At an equal ~1,300-token budget SAR-RAG wins (+0.056 recall, +0.116 complete-evidence). Against k=20 it loses on coverage. Both must be reported; budget-matched is the honest framing and is the strongest result the method currently has.
2	**Two named components are inert and one is near-inert.** Novelty and the similarity-gap rule change nothing at the frozen thresholds; the second representation is worth ~0.01 recall on 3.3% of queries.
3	**A trivial single-view baseline currently dominates the method.** Until this is refuted on held-out data, the central contribution is not established.

Outcome classification: (D) no convincing advantage — with a genuine but narrow **(C) efficiency-oriented trade-off at matched context budget**. Neither can be asserted as a result, because no confirmatory experiment has been run. The experiment is **not** invalid (E): no leakage, unfair prompt, broken metric or test-set tuning was found. It is unexecuted — and the screening evidence now predicts that the routing contribution, as currently framed, will not survive it.

10. Unresolved issues

#	Issue	Blocking
U1	No confirmatory run exists; all held-out results are PENDING	Yes
U2	Groq run-date rates unverified; cost undefined	Yes
U3	Complete-evidence recovery scored on 1,876 unanswerable MuSiQue examples	Yes
U4	Granularity confound: control added, not yet reported end to end	Yes
U5	Paper RRF formula does not match the implementation	Yes
U6	Second view: 16.7% of queries, unique gold in 3.3%, essentially QASPER-only	Yes
U13	**paragraph_only dominates the full routed system on every retrieval outcome**	Yes
U14	Novelty and similarity-gap components are provably inert at the frozen thresholds	Yes
U15	36.7% of queries terminate on a safety cap rather than adaptively	Yes
U16	Probe depth 5 beats frozen depth 3; must not be adopted without fresh calibration	No
U7	Non-inferiority margins likely unachievable; must not be revised post-hoc	No
U8	Failure tables and 4 of 9 figures need a real run	No
U9	Confirmatory figure hard-codes a 'pilot' caption	No

#	Issue	Blocking
U10	Revision pin inert on local snapshot path; provenance rests on verified sha256	No
U11	pytest not declared in pyproject.toml	No
U12	MuSiQue test labels hidden; test numbers require official submission	No